6.3 Medium Voltage (MV) Switchgear

6.3.1 Switchgear GIS/AIS Type

a. The preferable MV switchgear shall be Sulphur Hexafluoride (SF6) gas insulated switchgear (GIS) type and meet requirements in IEC 62271-203. If there is any project specific requirement for air insulated switchgear (AIS) type and meet requirements in IEC 62271-200; it shall be communicated with Micron Global Facilities in advance.

b. The MV switchgear should be rated for indoor environment.

c. The MV switchgear must be maintenance-free design with highest level of safety, performance and reliability.

d. The MV bus scheme shall be Single-Bus arrangement.

e. The MV switchgear shall apply *FAx-…-FA1-Main A-Tie-DS-Main B-FB1-…-FBy* configuration.

f. When extending either end of MV switchgear, it shall only be necessary to isolate the associated bus SF6 gas compartment and be able to remove end panel for the associated busbars. No other gas compartment and overall system operation shall be affected.

g. The primary components of the MV switchgear equipment such as busbar, vacuum circuit breaker, disconnecting switch, earthing/grounding switch, voltage transformer, current transformer shall be modularized and enclosed in grounded metal enclosure. Each primary component shall be constructed into separate gas cubicles with a pressure relief device.

h. The MV switchgear shall have arc resistant capability.

i. The internal components of MV switchgear shall be hermetically sealed with segregated single-phase enclosure provided.

j. MV switchgear technical data sheet is shown in the appendix.

6.3.2 Circuit breaker

a. The circuit breaker design, construction, testing and application shall satisfy ANSI/IEEE C37.04 and IEC 62271-100 as minimum requirement unless otherwise stated.
b. Circuit breaker shall be vacuum type (VCB) with highest switching duties, able to operate both electrically and mechanically as well as able to withstand any impacts during any harsh conditions.
c. Circuit breaker short circuit rating (making, interrupting, and braking) shall satisfy maximum prospective short circuit at present and future.
d. Circuit breaker short circuit capacity shall consider full DC component at present and future.
e. Required interlocks (w/ disconnecting switch, earthing/grounding switch, MV door compartment) shall be provided to prevent dangerous operations of circuit breaker.
f. Provide local manual close and trip function
g. Provide operating mechanism to automatically recharge the closing spring
h. Provide control power trip switch for disconnection of the control power circuit breaker.
i. Provide local-remote selector switch
j. Provide operation counter
k. Provide emergency machine off push button
l. Circuit breaker for transformer feeder shall be equipped with Inter-trip and close prevention interlocks w/ transformer door enclosure.
m. Medium voltage circuit breaker technical data sheet is shown in appendix.

6.3.3 Disconnecting and Earthing/Grounding Switch

a. Disconnecting/Circuit breaker and earthing/grounding switch shall be built-in three-position (Closed-Open-Earth/Ground) switch type. It shall have highest switching duties, able to operate both electrically and mechanically as well as able to withstand any impacts during any harsh conditions.
b. Mechanically and electrically block access to three-position switch operation during circuit breaker is in the closed position to prevent dangerous operations.
c. Provide dedicated operation handle for changing Closed-to-Open or Open-to-Closed
d. Provide dedicated operation handle for changing Open-to-Earth/Ground or Earth/Ground-to-Open

e. All operations status shall be monitored and indicated on the MV door compartment and Power SCADA.
f. Provide locking point of Earthing/Grounding switch
g. Provide visual verification window to allow actual visualization switching changes.
h. MV disconnecting and earthing/grounding switch technical data sheet is shown in appendix.

6.3.4 Protection and Metering

a. The protective relay shall be multi-function type with built-in advanced microprocessor unit and modular communication module.
b. The protective relay shall be able to operate continuously at minimum 10 years without any obsolete spare parts.
c. For Incoming portion need to be equipped with power quality grade metering device. The metering shall be power quality measurement grade with capability of power quality analysis function. Multi-function relays shall be provided as per clause 6.2.5 (c)
d. All protection devices and metering shall be provided communication protocol via IEC 61850 and linked through single-mode fiber optic network.
e. Transformer differential protection function (87T) shall be provided if applicable.
f. Line differential protection (87) shall be provided, if applicable.
g. Undervoltage relay (27) shall be provided.
h. Overcurrent phase fault relay with Instantaneous elements (50/51) shall be provided.
i. Lockout relay (86) shall be provided.
j. Synchronizing relay function (25) shall be provided.
k. Provide IED that suitable for motor type load.
l. Provide IED that suitable for transformers (Oil-Immersed) type load.
m. Earth/Ground fault relay with Instantaneous elements (50N/51N) shall be provided.
n. Provide CT shorting blocks for relay maintenance purpose.
o. If required, inter-trip and interlocking scheme for various protection shall be provided.
p. MV protection and metering devices technical data sheet is shown in appendix.

q. All protective devices shall be capable of handling Inrush currents under all service conditions, without abnormal ageing or deterioration.

6.3.5 Voltage Transformer (VT) and Current Transformer (CT)

a. VT

- The VT rating and location shall satisfy ANSI/IEEE C57.13 and IEC 60044-2 as minimum requirement unless otherwise stated. Rated output shall match the maximum load of the equipment connected and a future capacity of 25%. Secondary voltage of transformer shall preferably be 110V.
- VT shall be inductive single-phase busbar type mounted outside or inside metal enclosure with SF6 gas insulation. VT's compartment shall be fully segregated from other primary compartments of the MV switchgear.
- The VT rating (Frequency, ratio, ampere, accuracy, insulation level, temperature rise) shall meet system design requirement. Voltage transformers for measuring purposes shall be of accuracy class 1 for third parties (Utility) supply, Class 3 for internal supply and Class 0.5 for use with generator AVRs.
- VT shall be equipped with three-position switch (Close-Open-Earth/Ground).
- VT primary leads shall be protected by current limiting fuse which located in same VT gas-filled enclosure. VT for use with generator AVRs shall not be protected by fuses but shall be connected within the protection zone of generator differential protection. VT secondary leads shall be protected by MCCB and connected to low voltage compartment. All connections with in the low voltage compartment shall be solderless connectors on suitable terminal block.

b. CT

- The CT rating and location shall satisfy ANSI/IEEE C57.13 and IEC 60044-1 as minimum requirement unless otherwise stated.
- CT shall be toroidal type mounted outside or inside metal enclosure with full insulation between the primary conductor and current transformer core and secondary parts shall be provided by the SF6 gas insulation.
- The CT rating (Frequency, Ratio, Ampere, Accuracy, Insulation Level, Temperature Rise) shall meet system

design requirement. Dedicated CT shall be provided for Protection relaying and Metering. CT for measuring purposes shall be accuracy class 3 except those for measuring the supply from third parties (Utility), which shall be of class 1.

- Ratio changing of CT shall be possible by means of taps on the secondary windings (Min. 2 Taps).
- All secondary and tap leads of CT shall be connected to integral CT lugs inside the low voltage compartment. All connections within low voltage compartment shall be solderless connectors on suitable terminal block and no slide-on connection acceptable. CT shorting blocks shall be provided for maintenance or troubleshooting purpose.
- Current transformers shall have an appropriate VA rating and a saturation factor that will ensure the proper function of protective devices for all short-circuit currents up to the rated value of the switchgear.
- Where required for certain type of differential protection, the CTs shall be specified in terms of knee-point voltage, excitation current and secondary wiring resistance so as to ensure stability under through fault conditions.

6.3.6 Control power

a. Control power arrangement for closing, tripping and protection shall be either:
   - Combined DC supply unit/s with battery back-up for closing, tripping, protection and monitoring or
   - DC supply units with battery back-up for tripping and protection and AC supply for closing.
b. Provide disconnect breaker at least two separate DC power supplies with battery back-up feeders on each MV switchgear section. DC power supply shall be monitored for failure. A maintenance/failure selector switch, make-before-break, and blocking diodes for each DC source shall be supplied at the bus section/coupler.
c. MV switchgear control power shall be supplied via dedicated DC power supply which incorporates redundant chargers and redundant battery system with more than two strings. The charger and battery capacity shall have the capability of sustaining the worst operating conditions.
d. Output voltage ripple of DC supply shall not cause interference with the other equipment of the assembly.

e. Closing and tripping of switching devices shall not result in a voltage of less than the permissible voltage specified for the installed protective relays.

f. Preferred tripping and protection voltage supply shall be 110V DC. Each supply unit shall be capable of closing or charging at least two switching devices simultaneously and all others in succession per assembly. In addition, battery of each supply unit shall be capable of supply the entire assembly control circuit load which include electronic relays supply for a minimum period of 8 hours and then able to trip all connected switching devices in succession.

g. Closing/switching voltages for circuit breakers and disconnector/earthing (grounding) switch shall be 230V AC or 110V DC in combination with tripping and protection supply units.

h. Two separate 230V AC source should be provide to switchgear. Selector switch (break before make) at the bus coupler should allow change-over to alternate feed, if one supply is unavailable or taken out for maintenance.

i. Switchgear manufacturer shall provide high endurance (mechanical and electrical) and di-electric strength relays (include sockets), for final operational elements. This is to prevent abnormal operation due to faulty relays or sockets.

6.3.7 Automatic and Manual transfer operation

a. MV switchgear shall be equipped control logic that allow both automatic transfer system (ATS) and manual transfer system (MTS).

b. Provide Selector switch to choose between ATS or MTS at Tie Cubicle.

c. Conditions taken into consideration when design ATS and MTS for MV Switchgear.

   - If MV switchgear bus coupler (Tie) is always in closed position (NC), then ATS function is not required. MTS is still required.
   - If MV switchgear bus coupler (Tie) is always in open position (NO) and LV switchgear bus coupler is equipped with ATS function, then MV switchgear only require MTS.
   - If MV switchgear bus coupler (Tie) is always in open position (NO) and LV switchgear bus coupler is NOT equipped with ATS function, then MV switchgear require both function ATS and MTS.

d. MV switchgear shall be able to perform Auto-Transfer and Auto-Restore during testing and troubleshooting when ATS function is available.

e. Automatic transfer system operation shall meet all criterion prior to initiation.

- Both Incoming circuit breakers (NC), tie circuit breaker (NO), and disconnecting switch (NC) are in normal position, in-service, healthy, and disconnected from earthing/grounding switch.
- Power outage occurs at one side of Incoming line. This must be confirmed by at least two detections of undervoltage relays (27) at different point of measurement (Incoming line voltage and Bus voltage). Another Incoming line must be healthy, normal voltage, and free from bus fault or any fault.
- After confirming Item 2 above, dead side Incoming line shall be opened. Associated dead bus must be health and free from any bus fault or any fault.
- After item 1~3 are met then load transfer can be initiated by closing tie circuit breaker.

f. Automatic restore operation shall NOT be allowed. Only Manual Restoration is available.

g. Manual Restoration/Transfer Operation

System is normal with no protective function activated.

Outage supply line is healthy and ready to take load with line undervoltage relay (27) normalized.

6.3.8 Other accessories (Online Partial Discharge Monitoring, Surge Protection Devices, etc)

a. Gas Leakage detection

- An effective gas leakage detection system includes gauges, shall be provided for each compartment and connected to an alarm in Local control panel, Remote control panel, and Power SCADA/IMS.

b. Cable Bushing

- The cable bushing between HV switchgear feeder and primary of main transformer shall be supplied by HV switchgear vendor. The cable bushing should have separate oil and air insulated sections and serve as direct connection between the cable and main transformer.
- The SF6 connection comprises a housing which appropriately adapted to the power cable, connecting electrode and fitting for any misalignment and vibration. The cable bushing gas space shall be combined and monitored with the gas compartment of the GIS module to which it is connected.

c. Cable Termination

- For power cable type and size of outgoings which GIS are connected; shall be shown and matched in design single line diagram. Power company feeder as Incoming power cables and terminations will be provided by approved and licensed local power company Contractor. The power cable from HV switchgear from feeders to main transformer shall be supplied by HV switchgear vendor/contractor. The connector shield rings, cable housing, mounting plated and necessary accessories required for the installation of all cable termination shall be provided by HV switchgear vendor/contractor. All details shall meet power company standard.
- Cable glands/docking plate and mounting for single core cables shall be non-magnetic material and insulated from framework and switchgear mounting plate.

d. Lightning Arrester

- The lightning arrester shall be station class, metal-oxide or zinc-oxide type and insulated with SF6 gas filled-housing. The rating of lighting arrested shall meet project specified requirement as shown in data sheet.

e. Online Partial Discharge Monitoring

- To detect early stage of partial discharge inside the MV switchgear SF6 filled-compartment, online partial discharge monitoring system (Online PDMS) is required to be installed. UHF-based detection technology or equivalent or higher is recommended. Comprehensive partial discharge analysis shall be done automatically and provide clear alarm/warning signal to integrating power monitoring system (IMS). The diagnostic capability of Online PDMS shall compliant with IEEE Std. 1434 and IEC 60270.

f. Surge Protective Devices

- Surge Protective Devices shall prevent electrical equipment from over-voltage transient events by blocking or divert surge current to ground instead of damaging the electrical equipment.
- Surge Protective Devices shall be connected to line side of equipment.
- Selection of Surge Protective Devices to ensure equipment subjected to momentary over-voltage and duration is within the equipment withstand. The remaining energy is release to ground.

g. Blind cover

- The blind cover for each end shall be furnished. The covers will be used for blinding compartment during maintenance works.

h. SF6 Gas Filling Device

- One set of gas filling device including pressure reducing valve, manometer, adapter and filling hose shall be furnished for filling SF6 gas.

6.3.9 Installation

6.3.9.1 Transport, Storage and Erection

Assemblies supplied in transport units shall have these units clearly marked to facilitate assembly at site

Instruction for transport, storage and erection of the equipment shall be supplied by the Manufacturer as an integral part of the order.

Special tools and equipment required for erection, commissioning and maintenance shall form part of the order and shall be shipped together with the assembly.

6.3.9.2 General

Where necessary, owner or the Contractor shall submit Manufacturers' test reports, equipment certificates and site test reports, etc., for approval by the local authorities.

Unless specified, the minimum ingress protection for equipment shall be IP54 for outdoor installed equipment, IP41 for equipment installed indoors and IP2x for exposed parts of the equipment that has to be accessible during normal operation.

Foundation drawings, together with information on mass and loading data, shall be provided by manufacturer of Switchgear, to provide correct civil design of the relevant buildings and foundations. This include switchgear foundations, any inserts to be cast in, shall be within manufacturer's specific tolerances.

Switchgear arrangement and layout drawings shall be obtained from the manufacturer so that Contractor can incorporate into the overall arrangement drawings and cabling design.

Switchgear shall be installed so that there is sufficient space, as defined by manufacturer, to facilitate operation, maintenance and constructability requirement. Refer to clause 6.1 (f).

Switchgear shall only be installed when the switch room/substation civil and building works are completed, so as to minimize the ingress of dust/dirt and moisture during or after erection.

Substation floors shall be smooth and level to permit the handling of equipment on rollers, regardless of crane facilities provided.

Concrete floors shall have a surface protection layer to avoid formation/creation of dust.

MV Switchgear shall be located in a separate room from the controls, protection and metering room.

An equipment/switchgear access door should lead directly to the outside for each switch room.

6.3.9.3 Site Installation

Switchgear shall be installed in accordance with Manufacturer's installation instruction and supporting drawings.

SF6 switchgear shall be supervised and installed by the manufacturer competent person and/or manufacturer's trained/qualified persons. AIS switchgear shall at least be supervised by an experience qualified person/s familiar with the switchgear or by the manufacturer.

Switchgear modifications required during installation and/or commissioning shall be approved by the Owner prior to implementation. All modifications shall be recorded and information incorporated in the "as-built" drawings.

A site test certificate shall be issued by the manufacturer to confirm satisfactory completion of the recommended site test.